Distance matrices and weights

For each $\ell = 1, \ldots, m$: distance matrix $\mathbf{\Delta}_\ell$

Weight vector $\mathbf{w}$ with $\mathbf{1}^\top \mathbf{w} = 1$

Weighted and standardized Gram matrices

$\mathbf{J_w} = \mathbf{I}_n - \mathbf{1} \cdot \mathbf{w}^\top, \quad \mathbf{D_w} = \mathrm{diag}(\mathbf{w})$

Compute $\mathbf{G}_{\mathbf{w},\ell} = -\frac{1}{2}\mathbf{J_w}\mathbf{\Delta}_\ell\mathbf{J}_\mathbf{w}^\top$

Standardize $\mathbf{F}_{\mathbf{w},\ell} = \mathbf{D}_\mathbf{w}^{1/2}\mathbf{G}_{\mathbf{w},\ell}\mathbf{D}_\mathbf{w}^{1/2}$

Joint standardized Gram matrix

$$\mathbf{F}_\mathbf{w}^J = \sum_{\ell=1}^{m}\mathbf{F}_{\mathbf{w},\ell} - \frac{1}{m}\sum_{\substack{\ell,r=1\\ \ell\neq r}}^{m}\mathbf{F}_{\mathbf{w},\ell}^{1/2}\mathbf{F}_{\mathbf{w},r}^{1/2}$$

Final hybrid distance

$\mathbf{G}_\mathbf{w}^J = \mathbf{D}_\mathbf{w}^{-1/2}\mathbf{F}_\mathbf{w}^J\mathbf{D}_\mathbf{w}^{-1/2}$

$\mathbf{\Delta} = \mathbf{1}\mathbf{g}_\mathbf{w}^{J\top} + \mathbf{g}_\mathbf{w}^J\mathbf{1}^\top - 2\mathbf{G}_\mathbf{w}^J$